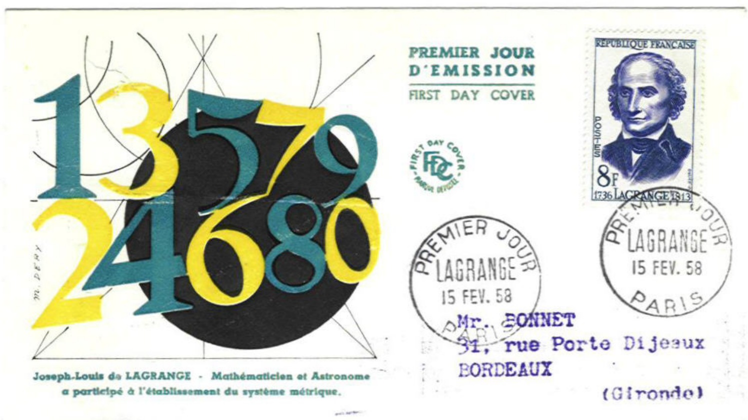
Quizz Week 10

Lagrange's Theorem



Cosets. Lagrange's Theorem.

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1. Consider the subgroup $H = \{0, \pm 3, \pm 6, \pm 9, \ldots\}$ of $\mathbb{Z}$. Mark the correct answers:
   (a) $11 + H = 17 + H$.
   (b) $-1 + H \neq 5 + H$.
   (c) $7 + H = 23 + H$.

Solution: We will use the following property of cosets:

$$a + H = b + H \Leftrightarrow a - b \in H$$

(a) is correct $11 + H = 17 + H$ since $11 - 17 = -6 \in H$.

(b) is not correct because $-1 + H = 5 + H$ since $-1 - 5 = -6 \in H$.

(c) is not correct because $7 + H \neq 23 + H$ since $7 - 23 = -16 \notin H$.

2. Using Lagrange's Theorem, determine whether the following statements are true or false:
   (1) Let G be a group and H and K subgroups of G with $|H| = 12$ and $|K| = 35$. Then $|H \cap K| > 2$. Can you determine $H \cap K$?
   (2) Every proper subgroup of a group of order 21 is cyclic.

Solution: (1) is **False:** Since $H \cap K$ is a subgroup of both H and K, *Lagrange's Theorem* implies that $|H \cap K|$ must divide 12 and 35. Thus $|H \cap K| = 1$, and so $H \cap K = \{1\}$.

(2) is **True:** Let G be a group of order 21. By *Lagrange's Theorem*, the only possible orders for the proper subgroups of G are 1, 3 and the 7. But a corollary of *Lagrange's Theorem* tells us that groups of prime order are cyclic, so the subgroups of G of order 3 and 7 are cyclic, and the subgroup of order 1 is the trivial subgroup $\{1\}$, which is also cyclic, since $\{1\} = \langle 1 \rangle$.